

RAJESH KANADE

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PROFESSIONAL SUMMARY

Generative AI Engineer with 1+ year of production experience designing, developing, and deploying LLM applications, RAG pipelines, and agentic AI systems. Skilled in LangChain, LangGraph, LlamaIndex, OpenAI, Anthropic, Gemini APIs, FastAPI, and Docker-based microservices. Experienced in fine-tuning (LoRA/QLoRA), prompt engineering, structured outputs (Pydantic), and semantic/hybrid search with FAISS, ChromaDB, and Pinecone. Proficient in LLM evaluation (DeepEval, Ragas), observability (LangSmith, LangFuse), CI/CD pipelines, and Agile/Scrum. Co-authored a BERT-based NLP research paper.

WORK EXPERIENCE

Generative AI Engineer / Full Stack Developer

Studium Tech Pvt. Ltd. | Pune | June 2025 – Present

- Designed and deployed production-grade LLM pipelines using LangChain, RAG, and OpenAI/Gemini APIs integrated into enterprise SaaS workflows serving thousands of users.
- Built agentic AI workflows with LangChain tool-calling and agent loops for multi-step reasoning and dynamic document search — aligned to AutoGen/CrewAI patterns.
- Implemented RAG architectures (chunking → embedding → vector store → retrieval) with ChromaDB, achieving sub-2s query response time and ~40% improvement in answer relevance.
- Engineered prompt versioning and structured output strategies (Pydantic, JSON Schema) to reduce LLM hallucinations ~30% and standardize downstream parsing.
- Built scalable async FastAPI and Node.js microservices for AI/LLM API integration into multi-tenant SaaS systems with JWT auth, RBAC, and Docker containerization.
- Integrated LangSmith and LangFuse for LLM observability, token/cost tracking, and pipeline debugging; collaborated in Agile/Scrum teams with CI/CD pipelines (GitHub Actions) and Git version control.

AI Training Data Annotator (Freelance / Remote)

Innodata Inc. | Dec 2025 – Present — RLHF Annotation | Prompt Engineering | Multimodal Data | LLM Alignment

- Contributed to multimodal AI model training pipelines via structured data annotation, including image inpainting prompt generation for diffusion model fine-tuning.
- Performed RLHF preference annotation — evaluated and ranked model-generated response pairs across accuracy, helpfulness, and safety to support LLM alignment workflows.
- Authored precise image reconstruction and object-removal prompts, and extracted visual pattern prompts from clothing/texture images to support dataset diversity for generative models.

TECHNICAL SKILLS

GenAI / LLMs: LangChain, LangGraph, LlamaIndex, OpenAI API, Anthropic API, Gemini API, Azure OpenAI, Hugging Face, Ollama, Prompt Engineering, Prompt Versioning, Structured Output (Pydantic, JSON Schema), Function Calling, Tool Use, Agentic Workflows

RAG & Vector DBs: RAG Pipeline Design, Chunking, Embeddings, Semantic Search, BM25 Hybrid Search, Re-ranking, FAISS, ChromaDB, Pinecone, pgvector, Weaviate

Agentic Frameworks: LangChain Agents, LangGraph (StateGraph, Conditional Edges, Tool Nodes, Checkpointing, Human-in-the-Loop), AutoGen, CrewAI, MCP Server/Client, Supervisor Patterns, Multi-Agent Orchestration

Fine-Tuning: LoRA, QLoRA, Hugging Face PEFT, SFT (Supervised Fine-Tuning), Instruction Tuning, Transformer Internals

Evaluation & Guardrails: DeepEval, Ragas, LangSmith, LangFuse, Guardrails AI, NeMo Guardrails, Custom Validators, Token & Cost Observability

Backend & APIs: Python, FastAPI (Async), Node.js, Fastify, REST APIs, gRPC, Microservices, JWT Authentication, RBAC

Cloud & DevOps: Docker, AWS S3, Azure Services, GCP Services, Git, GitHub Actions, GitOps/FluxCD, CI/CD, PM2

Databases: PostgreSQL, MongoDB, MySQL, Redis (Caching, Pub/Sub)

Frontend: React.js, Next.js, TypeScript, Material UI, Tailwind CSS, Streamlit

Domain: Domain-Driven Design (DDD), Multi-tenant SaaS, NBA/NAAC Accreditation Systems, Enterprise Integrations (Jira, GitHub)

GENERATIVE AI PROJECTS

AI Web Page Assistant & Video Explainer — Chrome Extension (PageMind)

Tech Stack: Python, LangChain, LangGraph, OpenAI API, FastAPI, ChromaDB, React.js, Pydantic | GitHub: github.com/rajeshkanade/pagemind

- Designed an end-to-end RAG pipeline with ChromaDB for webpage/video transcript ingestion, improving answer relevance ~40% vs. direct LLM calls.

- Applied prompt versioning and Pydantic-enforced structured output, reducing hallucinations ~30% and enabling reliable downstream parsing.
- Integrated OpenAI API for real-time summarization, contextual Q&A, and multi-modal content analysis (text + video transcripts), with LangSmith tracing for token-usage dashboards and latency profiling.

Multi-Modal Document Intelligence Platform

Tech Stack: Python, LangChain, LlamaIndex, AutoGen, OpenAI API, FastAPI, ChromaDB, Redis, Docker, Streamlit | GitHub: github.com/rajeshkanade/doc-intelligence

- Implemented hybrid search (semantic + BM25 + re-ranking) over ChromaDB with Redis caching, reducing query latency ~45% at enterprise scale.
- Built an agentic reasoning loop (AutoGen-style) for multi-document cross-referencing and intelligent summarization across 100+ page documents; containerized the full stack with Docker.
- Implemented RBAC-secured document access with Guardrails AI validators and a Streamlit UI, and integrated DeepEval for automated RAG evaluation — achieving a 0.87 faithfulness score with precision/recall scoring.

Accreditation Insight Generator — LangGraph Agentic Pipeline

Tech Stack: Python, LangGraph, LangChain, OpenAI API, FastAPI, PostgreSQL, MongoDB, Pydantic | GitHub: github.com/rajeshkanade/accreditation-ai

- Architected a multi-node LangGraph StateGraph pipeline with conditional edges — a Planner agent breaks an NBA/NAAC SAR into C1–C7 criteria, sub-agents process each criterion in parallel, and a Critic node validates compliance before final report generation.
- Built an LLM-driven gap-analysis system over structured institutional data (CO-PO mappings, CAY/CAYm1/CAYm2 metrics), improving reporting efficiency ~70% and cutting manual analysis time ~60%.
- Implemented a human-in-the-loop approval node (interrupt_before) and LangGraph MemorySaver checkpointing for stateful multi-session workflows, with Pydantic schemas for consistent JSON output to a React UI.
- Integrated PostgreSQL and MongoDB via custom MCP-style tool connectors in LangChain, enabling real-time institutional data queries within agent steps.

RESEARCH & PUBLICATIONS

BERT-Based Fake News Detection Using NLP — Co-authored under Prof. Kalpana Joshi, Fergusson College (Autonomous), Pune. Applied a BERT transformer for fake news classification, achieving high accuracy on benchmark datasets. (2025)

EDUCATION

Master of Computer Applications (MCA) — Fergusson College, Pune University | CGPA: 9.10 | 2024–2026

Bachelor of Science (Computer Science) — Savitribai Phule Pune University | CGPA: 8.87 | 2021–2024

CERTIFICATIONS & LANGUAGES

Certifications: DeepLearning.AI — LangChain for LLM Application Development | DeepLearning.AI — Building Systems with the ChatGPT API | Python Developer (Udemy) | JavaScript Intermediate (HackerRank)

Languages: English (Professional) | Hindi (Native) | Marathi (Native)